

Jordan University of Science and Technology
Department of Mathematics and Statistics
Exam 1 (Math 103)-Fall 2024

14
30

الاسم: _____ الرقم الجامعي: _____ الشعبة: 1
وقت المحاضرة: _____ اسم المدرس: _____ الرقم المتسلسل: 18

This exam consists of 15 multiple choice questions (2 points each). Put the correct answer in the table.

Question#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answers	<u>A</u>	<u>C</u>	<u>B</u>	<u>A</u>	<u>C</u>	<u>C</u>	<u>A</u>	<u>C</u>	<u>A</u>	<u>B</u>	<u>D</u>	<u>D</u>	<u>B</u>	<u>C</u>	<u>B</u>

1. If $f'(x) = \cos(x) - 2$, where $f\left(\frac{\pi}{2}\right) = 3$. Then

- A) $f(x) = \sin(x) - 2x$ B) $f(x) = \sin(x) - 2x + \pi + 2$ C) $f(x) = \cos(x) + \pi + 2$ D) $f(x) = \sin(x) + \pi$

2. The integral $\int \frac{\cos(\sqrt{x})}{\sqrt{x}} dx =$

- A) $2 \sin(\sqrt{x}) + c$ B) $\sin(\sqrt{x}) + c$ C) $2 \cos(\sqrt{x}) + c$ D) $\cos(\sqrt{x}) + c$

3. The value of the integral $\int_0^1 \tan^{-1}(x) dx =$

- A) $\frac{\pi}{2} - \frac{\ln(2)}{2}$ B) $\frac{\pi}{4} + \frac{\ln(2)}{2}$ C) $\frac{\pi}{4} - \frac{\ln(2)}{2}$ D) $\frac{\pi}{4} + \frac{\ln(3)}{2}$

4. The value of the integral $\int_1^e \ln(x) dx =$

- A) $e - 1$ B) -1 C) e D) 1

5. The suitable partial fraction decomposition of $\frac{x^2+3}{(x^2-4x+4)(x^2+x-6)}$ is

A) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+2} + \frac{D}{x-3}$

~~X~~ B) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} + \frac{D}{x+3}$

C) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx+D}{(x^2+x-6)}$

~~X~~ D) $\frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{x-2} + \frac{D}{x+3}$

6. If $\int_0^1 (1+kx)e^x dx = e-6$, then the value of $k =$

A) 5

B) -10

C) 10

D) -5

7. The solution to the differential equation $\frac{dy}{dx} = xe^{x^2}$, $y(0) = 2$ is

A) $y(x) = \frac{1}{2}e^{x^2} + \frac{3}{2}$

~~X~~ B) $y(x) = e^{x^2} + \frac{1}{2}$

C) $y(x) = \frac{1}{2}e^x + \frac{3}{2}$

~~X~~ D) $y(x) = e^{x^2} - \frac{3}{2}$

8. The integral $\int x^2 \sin(x^3) dx$ is

A) $\frac{1}{3} \cos(x^3) + C$

B) $-\frac{1}{3} \sin(x^3) + C$

C) $-\frac{1}{3} \cos(x^3) + C$

D) $\cos(x^3) + C$

9. The particular solution to the separable differential equation $\frac{dy}{dx} = -y^2(3x^2+2x)$, $y(1) = 3$ is

A) $y(x) = \frac{1}{x^3+x^2-\frac{5}{3}}$

B) $y(x) = x^3+x^2-\frac{5}{3}$

C) $y(x) = \frac{1}{x^3+x^2-5}$

D) $y(x) = \frac{1}{x^3+x^2+\frac{5}{3}}$

10. The particular solution to the linear differential equation $(x^2+1)\frac{dy}{dx} + 2xy = x$, $y(0) = 2$ is:

A) $y(x) = \frac{x^2+2}{(x^2+1)}$

B) $y(x) = \frac{x^2}{2(x^2+1)}$

C) $y(x) = \frac{x^2+4}{2(x^2+1)}$

D) $y(x) = \frac{x+3}{2(x^2+1)}$

11. The integrating factor $\mu(x)$ for the linear differential equation $x^3 y' - 3x^2 y = x^6 e^x$, $x > 0$ is:

A) $\mu(x) = x^3$

~~B) $\mu(x) = e^{-x^3}$~~

~~C) $\mu(x) = e^{x^3}$~~

D) $\mu(x) = \frac{1}{x^3}$

12. The integral $\int \frac{\sin(x)}{\cos^2(x)} dx =$

A) $-\cos(x) + c$

B) $\sin^2(x) + c$

C) $-\sec(x) + c$

D) $\sec(x) + c$

13. Which one of the following substitutions will convert the Bernoulli differential equation $y y' + \frac{2}{x} y^2 = y^4 x$ into linear differential equation:

A) $v = y^2$

B) $v = y^{-2}$

C) $v = y^{-4}$

D) $v = y^{-3}$

14. Which one of the following $M(x, y)$ will make the differential equation: $M(x, y) dx + (2y + x^2 + 1) dy = 0$ an exact equation:

A) $M(x, y) = xy - 9x^2$

B) $M(x, y) = 12xy + x^2$

C) $M(x, y) = 2xy + x^2$

D) $M(x, y) = 3xy - 9x^2$

15. Which one of the following differential equations is nonhomogeneous:

A) $\frac{dy}{dx} = \frac{x + y^2}{x^2 + y^2}$

B) $(y^2 x + 3x^2 y) dx + (x^3 + y^3) dy = 0$

C) $y(\ln(y) - \ln(x) + 1) dx - x dy = 0$

D) $(3x^2 - y^2) dx + \left(xy - \frac{x^3}{y}\right) dy = 0$